
CSE3068 - SEQUENTIAL AND SPATIAL DATA MINING

DA-1

Flight Delay Propagation Analysis using Sequential and Spatial Data Mining

1. Problem Statement :

Normally delays occur in a cascade – when a flight is delayed at one airport, it is carried to the airport on the downstream side of the journey, or other flights which have a similar aircraft and crew route to and from the airport (e.g. Delhi to Mumbai and then to Bangalore). It is important both the airline companies and passengers are aware of what combinations of airports (and routes) have been found to experience cascading delays and what regions (airport clusters, hub zones) are found to be “delay hotspots” to help anticipate and limit disruptions.

Each of the traditional approaches (regression or single flight classifiers) that have been used to predict delays in the past treats one flight at a time, but they overlook two important aspects of this course:

Sequential Structure: How the flight sequence of delays is propagated over a multiple-leg flight.

Spatial Structure: Airport spatial characteristics - airport density, etc. among airport, regional weather zones, etc that impact the distribution of airport delays.

With the data being a set of historical flight operations (flight number, origin, destination, scheduled/real times, delay minutes in flight time, airport coordinates) the problem is to obtain:

Multiple successive time delays on multi leg path. Create airports/routes that introduce spatial clusters and organise them. Generate spatially-correlated clusters of airports/routes with delays. The flights are chained together by spatio-sequential rules which are likely to link the two flights: If a flight leaves the airport from the congested-cluster, then it is likely that the next flight leaving that airport will be delayed.

2. Overview of the Solution

Step 1: Data Preparation

Please download either the U.S. DOT Bureau of Transportation Statistics (BTS) On-Time Performance Dataset or the Kaggle dataset “2015 Flight Delays and Cancellations.”

Wclean the data set removing the inconsistencies, if any, and information missing from the set. Generate data chain segments of flights on per-aircraft/per-route basis.

Transform flight data to ordered sequences of flights per tail and/or per daily route of each air craft.Include airport coordinates [latitude, longitude] in a spatial analysis.

Step 2: Sequential Pattern Mining

For each aircraft trigger a symbolic sequence with its journey.

Apply Pattern-Growth, DFS algorithm – PrefixSpan.Compare with GSP - (Generalized Sequential Pattern) (Apriori like, Breadth-First Search).

Discover:

- Properly propagate delays.
- Common delay patterns
- Delay chain behaviors

Use Edit Distance and Alignment-based Distance to sort similar Delay Sequences.

Step 3: Spatial Data Mining

Create a spatial network for an airport from:

- Airport coordinates
- Flight route connections

Use spatial clustering algorithms:

- DBSCAN
- ST-DBSCAN (Space-Time DBSCAN)

Step 4: Spatio-Sequential Integration

Ensure that the sequential patterns mined from the data and spatial cluster information are included.

Understand propagation of delays in spatial clusters (according to integrated rules).

Make a Sketch of the airport scheme using:

NetworkX (Graph Representation)

Folium (Interactive Maps)

Represent:

- Airports as Nodes
- Directed Edges are used to represent the flight routes.

Employ 'Delay' sequences with Colors.

Step 5: Evaluation

Analyze the patterns of a Mining Operation by:

Sequential Pattern Evaluation

- Support
- Confidence

Spatial Clustering Evaluation

- Silhouette Score
- Prediction Model (Extension)

Develop a model to modelize delays using:

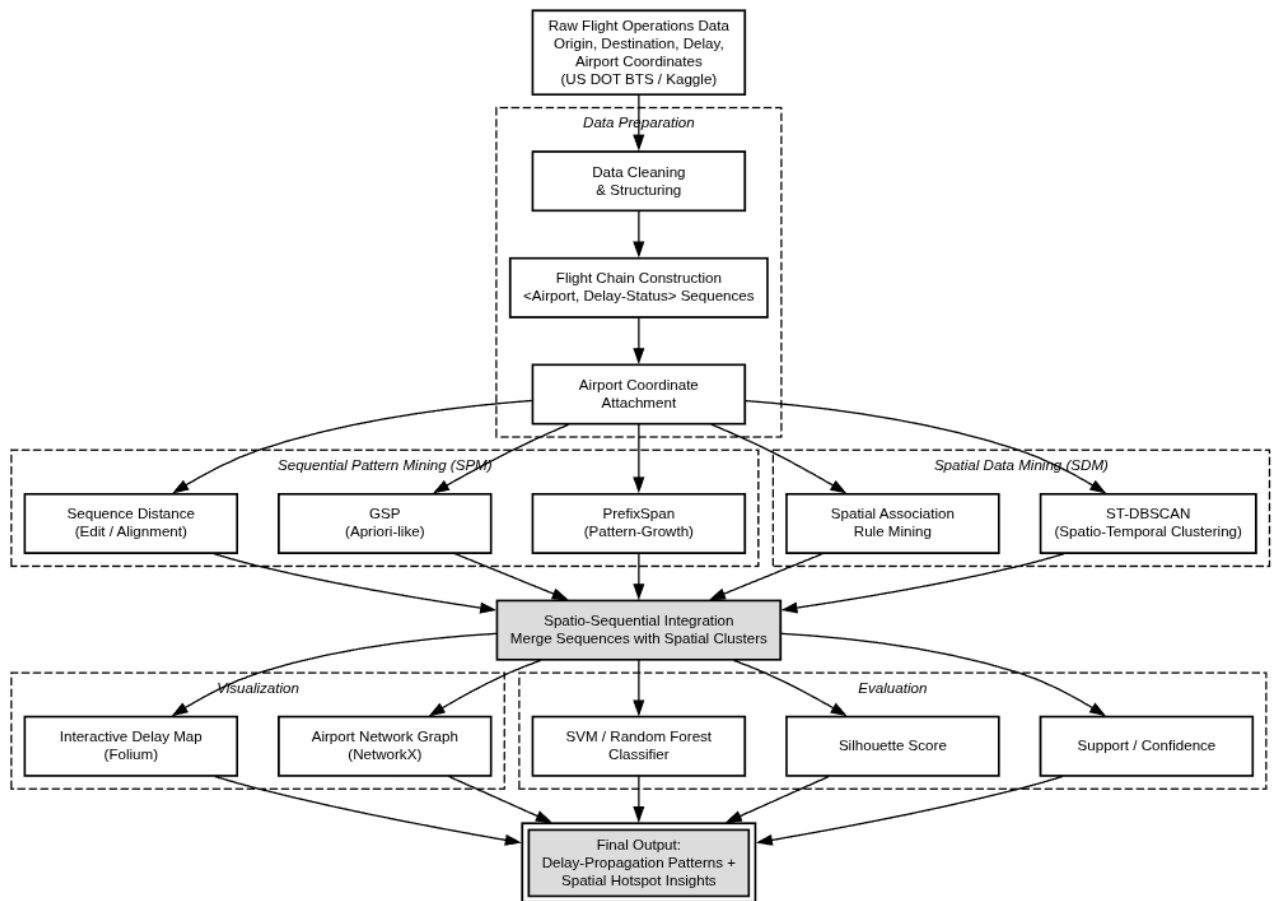
- support vector machine (SVM).
- Random Forest

Compare its performance to the conventional single-flight prediction techniques.

3. Tools and Technologies Used

Category	Tools / Technologies
Programming Language	Python
Sequential Pattern Mining	PrefixSpan, mlxtend (GSP/Apriori)
Spatial Data Mining	Scikit-learn (DBSCAN), ST-DBSCAN, GeoPandas
Graph Representation	NetworkX
Visualization	Matplotlib, Folium, Plotly
Data Processing	Pandas, NumPy
Dataset	US DOT BTS On-Time Performance Dataset / Kaggle "2015 Flight Delays and Cancellations" Dataset

4.Workflow :



DATASET LINK -

<https://www.kaggle.com/daryaheyko/airline-on-time-statistics-and-delay-causes-bts>